

CPSC 471 Project – EERD

Group 32

February 2026

1 Overview

This project models an internal merchandise ordering system where employees place orders for company merchandise. Admin and SuperAdmin users are responsible for managing catalog and operational workflows. The design supports essential components of an ordering system including product categorization, product images, product variants (size/color combinations), ordering via order lines, payment records, and email notifications.

2 EERD Diagram

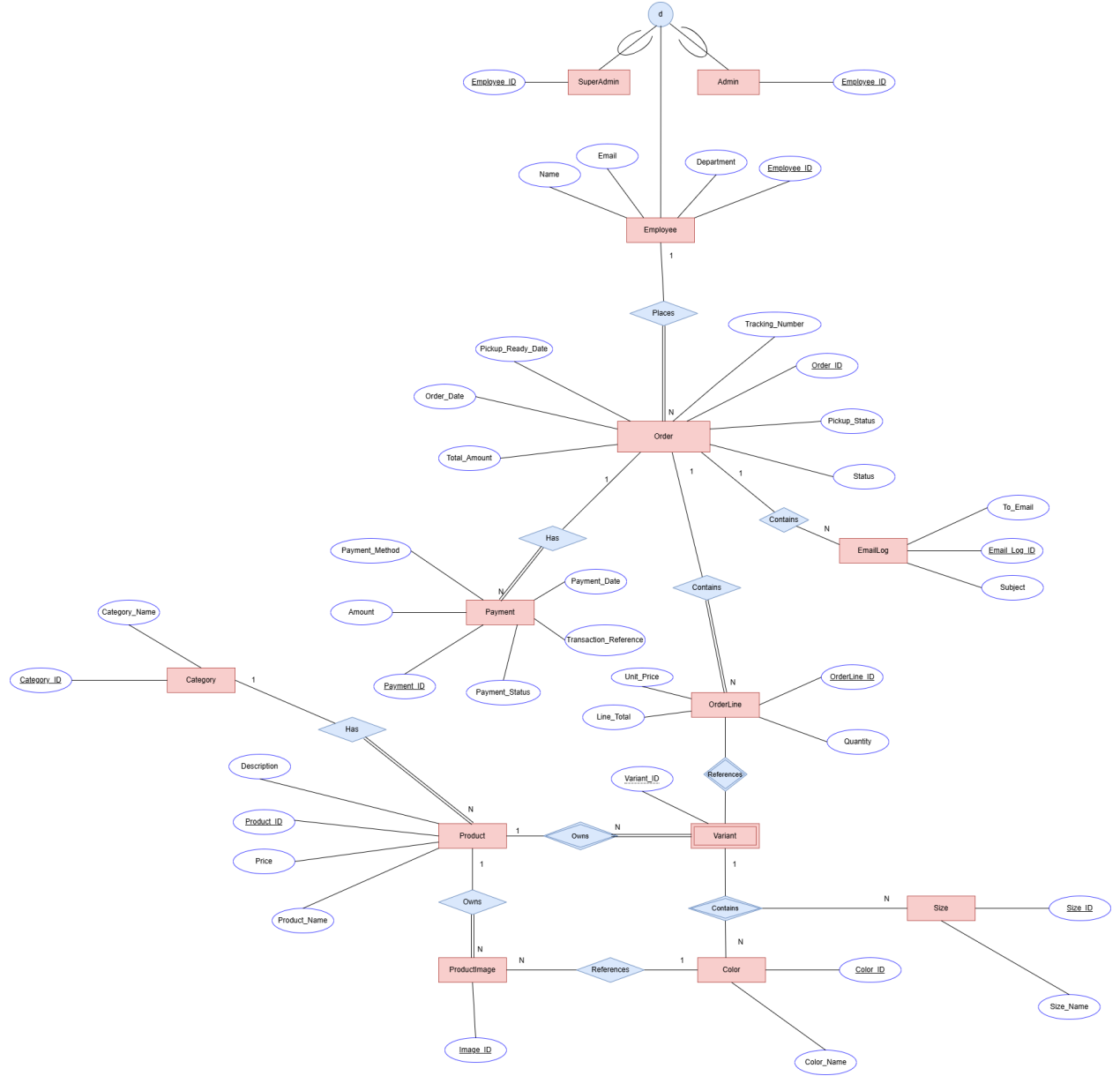


Figure 1: Enhanced Entity-Relationship Diagram (EERD) for the BrightCore Merch Store system.

3 Key Design Assumptions

3.1 Users and Roles (Specialization)

- **Employee** is the supertype representing any user who can interact with the system.
- **Admin** and **SuperAdmin** are subtypes of **Employee** with elevated privileges.
- The specialization between **Admin** and **SuperAdmin** is **disjoint**: one **Employee** cannot be both an **Admin** and a **SuperAdmin**.

- The specialization is **partial**: some Employees may be neither Admin nor SuperAdmin (such as regular employees who place orders).

3.2 Categories and Products

- Each Product belongs to exactly one Category.
- A Category may contain zero or many Products.
- Each Product may have zero or many ProductImages.
- Each ProductImage belongs to exactly one Product.
- A ProductImage may optionally reference a Color (used when an image is color-specific); otherwise the Color association is absent.

3.3 Variants (Weak Entity)

- **Variant** is modeled as a **weak entity** owned by Product.
- The relationship *Product identifies Variant* is an **identifying relationship**.
- Every Variant must be associated with exactly one Product (total participation of Variant in the identifying relationship).
- Each Product may have zero or many Variants.
- Each Variant references exactly one Size and exactly one Color in the current EERD. If needed, this can be relaxed to optional Size/Color to support one-size or colorless items.

3.4 Orders and Order Lines

- Each Order is placed by exactly one Employee (total participation of Order in the Places relationship).
- Each Employee may place zero or many Orders.
- Each Order contains one or many OrderLines.
- Each OrderLine belongs to exactly one Order.
- Each OrderLine references exactly one Variant.
- A Variant may appear in zero or many OrderLines across multiple Orders.

3.5 Payments

- Each Payment belongs to exactly one Order (total participation of Payment in the Has relationship).
- An Order may have one or many Payments. This supports payment retries, split payments, or multiple transaction records.

3.6 Email Logs

- Each EmailLog record is associated with exactly one Order in the current EERD.
- An Order may have zero or many EmailLog records (e.g., confirmation email, pickup-ready email, cancellation email).

4 Relationship Summary (Cardinality and Participation)

The following summarizes the main relationship constraints shown in the EERD:

- Employee *Places* Order: Employee (0..N), Order (1..1)
- Order *Contains* OrderLine: Order (1..N), OrderLine (1..1)
- OrderLine *References* Variant: Variant (0..N), OrderLine (1..1)
- Category *Has* Product: Category (0..N), Product (1..1)
- Product *Owns* ProductImage: Product (0..N), ProductImage (1..1)
- Product *Identifies* Variant (weak entity): Product (0..N), Variant (1..1, total)
- Size *Applies to* Variant: Size (0..N), Variant (1..1)
- Color *Applies to* Variant: Color (0..N), Variant (1..1)
- Order *Has* Payment: Order (1..N), Payment (1..1)
- Order *Has* EmailLog: Order (0..N), EmailLog (1..1)
- Color *May be referenced by* ProductImage: Color (0..N), ProductImage (0..1)

5 Notes on Requirements

- The design contains more than eight unique entity types (e.g., Employee, Admin, SuperAdmin, Order, OrderLine, Product, Category, Variant, Size, Color, Payment, EmailLog, ProductImage).
- Variant is modeled as a weak entity type.
- The design includes more than ten relationship types across catalog, ordering, and logging.
- The design includes at least two distinct end-user types (Employee vs Admin/SuperAdmin).